

Def.

Let G be a group and A be a non-empty subset. A map

$$\alpha: G \times A \rightarrow A: (g, a) \mapsto g \cdot a$$

is called a group action of G on A if it satisfies

- 1) For all $a \in A$, $1_G \cdot a = a$
- 2) For all $g_1, g_2 \in G$, and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Prop. Given a group action α of G on A , there exists a homomorphism

$$\mathcal{Q}_\alpha: G \rightarrow S_A: g \mapsto \sigma_g \quad \text{where} \quad \sigma_g: A \rightarrow A: a \mapsto g \cdot a.$$

which is called the permutation representation associated to the given action.

Conversely, given a homomorphism $\mathcal{Q}: G \rightarrow S_A$, there exists a group action

$$\alpha_\mathcal{Q}: G \times A \rightarrow A: (g, a) \mapsto \mathcal{Q}(g)(a)$$

Proof. We only need to check that the given map is well-defined.
Let $g \in G$ be fixed. Then, $\sigma_g \circ \sigma_{g^{-1}}(a) = \sigma_{g^{-1}g}(a) = a$, therefore $\sigma_g \in S_A$.
The converse is obtained by the definition of group action.

Def. Let G acts on A . For each $a \in A$, a subset

$$\text{Orb}_G(a) \stackrel{\text{def}}{=} \{g \cdot a \mid g \in G\} \subseteq A$$

is called the Orbit of a .

Prop. Let a group G acts on a set A . Given $a \in A$, a map

$$G/G_a \rightarrow \text{Orb}_G(a): gG_a \mapsto g \cdot a$$

is bijective. In particular, if G is finite, then

$$|G/G_a| = \#\text{Orb}_G(a).$$

Proof. Given $g_1, g_2 \in G$,

$$g_1 G_a = g_2 G_a \Leftrightarrow g_1^{-1} g_2 \in G_a \Leftrightarrow g_1^{-1} g_2 \cdot a = a \Leftrightarrow g_1 \cdot a = g_2 \cdot a$$

So it is well-defined and injective. And the surjectiveness is clear. $\square$

Thm.

Let G be a group and $H \leq G$ be a subgroup.

Consider G acts on $\{gH \mid g \in G\}$ by left multiplication, and affords homomorphism

$$\pi_H: G \rightarrow S_A: g \mapsto \sigma_g \quad \text{where } A = \{gH \mid g \in G\} \\ \text{and } \sigma_g(xH) = gxH.$$

Then G acts transitively on A and the stabilizer $G_{1 \cdot H} = H$, and

$$\ker \pi_H = \bigcap_{x \in G} xHx^{-1} \leq H$$

is the largest normal subgroup of G contained in H .

Proof. Given $aH, bH \in A$, $bH = ba^{-1}(aH)$, so aH and bH lie in same orbit.

And, since $gH = H \Leftrightarrow g \in H$,

$$G_{1 \cdot H} = \{g \in G \mid gH = H\} = H.$$

To show the last assertion, by definition,

$$\ker \pi_H = \bigcap_{x \in G} G_{xH}$$

$$\text{and } G_{xH} = \{g \in G \mid gxH = xH\} = \{g \in G \mid x^{-1}gx \in H\} = xHx^{-1}.$$

Moreover, $\ker \pi_H \trianglelefteq G$ because it is kernel of the homomorphism, and for any normal subgroup $N \trianglelefteq G$ of G contained in H ,

$$\text{for all } x \in G, \quad N = xNx^{-1} \leq xHx^{-1}$$

therefore $N \leq \ker \pi_H$. $\square$

Def. Let G be a group, and let $H \leq G$ be a subgroup.

A subgroup $\text{Core}_G(H) \stackrel{\text{def}}{=} \bigcap_{g \in G} gHg^{-1}$ is called the core of H in G .

Note: the theorem above gives

$$H \trianglelefteq G \Leftrightarrow H = \text{Core}_G(H).$$

Thm.

Let G be a finite group of order n , and let p be the smallest prime dividing n . Then, every subgroup of index p is normal.

Proof. Let $H \leq G$ with $|G:H| = p$. Let G acts on left cosets of H and

$$\pi_H: G \rightarrow S_p$$

be the afforded permutation of the given action.

Then, $\ker \pi_H \leq H$, let $K = \ker \pi_H$ and $R = |H:K|$. Then,

$$|G:K| = |G:H| \cdot |H:K| = pR.$$

Since $G/K \cong \text{Im } \pi_H \leq S_p$, so pR divides $p!$ by the Lagrange's Theorem.

Therefore, R divides $(p-1)!$. But all prime factors of $(p-1)!$ are less than p , by the minimality of p , R must be 1, that is, $H = K = \ker \pi_H \trianglelefteq G$ $\square$

Note: let G acts on finite set A which has m elements.

Then, the afforded permutation $\varphi: G \rightarrow S_m$ satisfies:

$$G/\ker \varphi \cong \text{Im } \varphi \leq S_m \text{ implies } |\text{Im } \varphi| \mid |G| \text{ and } |\text{Im } \varphi| \mid m!$$

So, if p is the smallest prime factor of $|G|$, then

$$m < p \Rightarrow \text{Im } \varphi = 1 \quad \text{and} \quad m = p \Rightarrow |\text{Im } \varphi| \in \{1, p\}.$$

In the case of the above theorem, the action is transitive, therefore $\text{Im } \pi_H \neq 1$. Therefore, $|\ker \pi_H| = |H|$ and $\ker \pi_H \leq H$, so proved.



